Twacha — Project Completion Analysis

Blueprint Phase Overview

Phase	Timeline	Status
Phase 1 — Schema & Contract	Week 1–2	■ ~85% Complete
Phase 2 — Recommender & Dietary Log	Week 3–5	■ ~60% Complete
Phase 3 — Vision Model (CV)	Week 3–8	■ 0% — Still mocked
Phase 4 — Integration & Polish	Week 7–10	■ 0% — Not started

What's Built & Working ■

Backend (Person A) — Solid Foundation

Component	File	Status	Notes
FastAPI scaffold	main.py	■ Working	v0.2.0, auto-creates tables on startup
Database layer	database.py	■ Working	SQLite dev, Postgres-ready via `DATABASE_URL`
SQLAlchemy models	models.py	■ Working	`SkinScan`, `Program`, `DietaryLog`, `NutrientEvidenceRule` — all 4 tables from blueprint
Pydantic schemas	schemas.py	■ Working	Matches §4 contract exactly — all enums (`EvidenceTier`, `Trend`, `AqiBand`, `DominantPollutant`)
10-nutrient scorer	recommender.py	■ Working	Published coefficients from Fabbrocini, Krutmann, Jung, Vaughn, etc. + AQI modifier
Food lookup + filtering	nutrient_lookup.py	■ Working	INDB + GI sourced, 10 nutrients × 4-8 foods each, dietary log filtering works
AQI client	aqi_client.py	■ Working	WAQI integration with auto mock fallback
Mock CV	mock_cv.py	■ Stub	Returns random plausible scores — placeholder for Phase 3
`POST /api/scan/`	scans.py	■ Tested	Full pipeline: CV → AQI → score → filter → trend → save → respond
`GET /api/history/`	scans.py	■ Tested	Returns program progress + scan list
`POST /api/log/`	scans.py	■ Tested	Upserts daily dietary log
`GET /api/log/`	scans.py	■ Tested	Returns last N days of logs
Trend calculation	recommender.py	■ Tested	Rolling avg comparison, 0.5 threshold, smoothed

Flutter (Person B) — ■■ NOT STARTED

> [!CAUTION] **No Flutter project exists anywhere in the workspace.** The blueprint expects 7 screens scaffolded in Phase 1. This is the single biggest gap.

What's Missing — Ordered by Priority

■ Critical (Blocking Progress)

1. Flutter App — Not Started at All

The blueprint §10 specifies **7 screens** that Person B should have scaffolded by end of Phase 1:

Screen	Purpose	API Calls
Consent / Onboarding	Privacy explanation, program start	None (local)
Home / Dashboard	Today's scores, AQI badge, top recs	`GET /api/history/`
Capture	Camera + face alignment overlay	None (pre-upload)
Result	Scores + recommendation cards	`POST /api/scan/`
Trend / History	7-day line chart, Day X of 30	`GET /api/history/`
Diet Log	User logs food	`POST /api/log/`, `GET /api/log/`
Recommendation Detail	Full explanation + source for one rec	(from Result state)

Stack specified in blueprint: Flutter + Provider (state management) + Dio (HTTP) + fl_chart (charts) + camera package.

2. Phase 3 CV Model — 0% Started

mock_cv.py still returns random values. The blueprint §6 specifies:

• **Acne**: MobileNetV2 or YOLOv8-Nano fine-tuned on ACNE04 dataset

• **Redness**: Classical CV — LAB/HSV color-space deviation (OpenCV)

• **Dryness**: Classical CV — texture roughness + specular highlight analysis (OpenCV)

• **ITA**: Individual Typology Angle from L*a*b* pixel values

> [!TIP] **Blueprint fallback plan**: Ship redness + dryness (OpenCV, days not weeks) + lesion-count heuristic for acne. Start here for an early demoable win.

■ Partially Done (Phase 2 Gaps)

3. `seed_rules.py` is Broken

seed_rules.py imports `RecommendationRule` and `SEED_RULES` — but the current codebase has `NutrientEvidenceRule` (not `RecommendationRule`) and `SEED_RULES` doesn't exist in recommender.py. The seeder script is outdated from v1 and will crash if run.

4. Escalation Logic — Not Built

Blueprint §5 says: *"If a nutrient target has been top-ranked for 7+ days and trend is* `flat`*, diversify to the next-ranked target."* This is a Phase 2 deliverable — not yet implemented.

5. Alembic Migrations — Not Set Up

`alembic` is in requirements.txt but `alembic init` has never been run. The app uses `Base.metadata.create_all()` on startup. The blueprint explicitly says: *"Set up* `alembic init` *before your schema stabilizes."*

6. Render Persistence Decision — Not Made

Blueprint §2 flags this as a **decide-now** item: SQLite wipes on every Render redeploy. Need to choose Render Postgres or Render Persistent Disk before Phase 4.

■ Minor Gaps

7. No `user_id` FK

`Program` and `SkinScan` are currently single-user (one program per database). Need `user_id` FK once accounts exist.

8. AQI Smoothing

Blueprint §9 says use a rolling 3-7 day AQI average. Current aqi_client.py fetches live AQI for each scan — no smoothing.

9. Trend Per Metric

Current scans.py computes trend for `acne_score` only. Blueprint §5 implies trend per metric — each recommendation should have its own metric-specific trend.

Recommended Next Steps (In Order)

Immediate — Do This Week

```
graph TD
    A["1. Fix seed_rules.py<br/>(30 min)"] --> B["2. Set up Alembic<br/>(1 hour)"]
    B --> C["3. Start Flutter scaffold<br/>(2-3 days)"]
    C --> D["4. Start OpenCV redness/dryness<br/>(2-3 days)"]
    D --> E["5. Wire Flutter to backend<br/>(1 week)"]
    E --> F["6. Train acne model on ACNE04<br/>(2-3 weeks)"]
```

#	Task	Who	Time	Details
1	**Fix** `seed_rules.py`	Person A	30 min	Update imports to `NutrientEvidenceRule`, seed from `EVIDENCE_FALLBACKS` in scans.py — or delete and rely on the existing fallback dict
2	**Run** `alembic init`	Person A	1 hour	Set up migration framework before adding more models
3	**Scaffold Flutter app**	Person B	2-3 days	`flutter create twacha_app`, add Dio + Provider + fl_chart + camera, scaffold all 7 screens with dummy data
4	**Build OpenCV redness + dryness + ITA**	Person A	2-3 days	Classical CV — no ML training needed. Replace the 3 functions in mock_cv.py. Ship this for an early demo.
5	**Wire Flutter to live backend**	Person B	1 week	Connect each screen to the corresponding API endpoint. Phase 1 is "done" when `POST /api/scan/` from Flutter renders correctly shaped scores on the Result screen.

Near-Term — Next 2-4 Weeks

#	Task	Who	Time
6	Train acne model (MobileNetV2/YOLOv8-Nano on ACNE04 in Colab)	Person A	2-3 weeks
7	Implement escalation logic (stale recommendation diversification)	Person A	2-3 hours
8	Add per-metric trend calculation (not just acne)	Person A	1 hour
9	Add AQI smoothing (rolling 3-7 day average)	Person A	1-2 hours
10	Consent screen + "not a diagnosis" disclaimers	Person B	1 day
11	Day 7/14/30 milestone screens	Person B	2 days
12	Evidence tier badge color coding (green/amber/terracotta)	Person B	half day

Pre-Deploy — Phase 4

#	Task
13	Decide Render Postgres vs Persistent Disk
14	Swap mock CV for real model in `mock_cv.py`
15	5-10 test users, full 30-day program
16	Evaluate: precision/recall on ACNE04 test split, user survey on recommendation relevance

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Summary Score

Area	Score	Comment
API Contract	**10/10**	Fully matches §4 — all 4 endpoints, all field types, all enums
Database Schema	**9/10**	All 4 models present. Missing `user_id` FK and Alembic.
Recommender Engine	**9/10**	Published coefficients, AQI modifier, dietary filtering — all working. Missing escalation logic.
Food Database	**8/10**	10 nutrients × 4-8 foods each, INDB+GI sourced. Could expand to 15 rules per blueprint §11.
CV Pipeline	**1/10**	Completely mocked — the biggest remaining technical risk.
Flutter App	**0/10**	Does not exist.
Privacy & Image Lifecycle	**10/10**	No image column, process-and-discard enforced by design.
Deployment Readiness	**2/10**	No Alembic, no persistence decision, no Render config.

> [!IMPORTANT] **The backend is genuinely well-built** — the API contract, recommender, and food database are production-quality. The two critical blockers are: (1) no Flutter frontend at all, and (2) mock CV model. Start the Flutter scaffold and OpenCV classical CV simultaneously — both are "days not weeks" work that unblocks everything downstream.